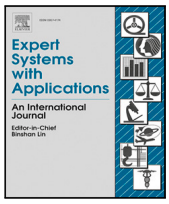
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MHCD: Multi-Hierarchy Interactive Constraint-Aware Cognitive Diagnosis Framework

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ABSTRACT

Cognitive Diagnosis (CD) is a fundamental aspect of educational data mining that focuses on assessing students' mastery of specific knowledge domains by analyzing their response patterns. However, traditional CD frameworks often fail to reasonably model students' true knowledge state due to insufficient constraints for automatic inference, particularly overlooking the hierarchical relationships among explicit behavioral records. This limitation undermines both the rationality and interpretability of diagnostic outcomes. To address these challenges, we propose an innovative ***M**ulti-**H**ierarchy Interactive Constraint-aware **C**ognitive **D**iagnosis (**MHCD**)* framework. This framework first identifies and constructs the hierarchical relationships of explicit records in real-world educational contexts. It then delves into feature interactions across multiple hierarchies and achieves more accurate and reasonable knowledge state modeling through an iterative constraint-aware strategy. Finally, it integrates multi-hierarchy interaction-aware features with personalized features, significantly enhancing the expressive power of both individual and hierarchical attributes. Experimental results demonstrate that MHCD achieves notable improvements in diagnostic accuracy, rationality, and interpretability, while seamlessly integrating into existing cognitive diagnosis frameworks to further enhance their overall performance.

1. Introduction

Cognitive Diagnosis (CD), as one of the key tasks in educational data mining, focuses on analyzing students' response records to gain insights into their mastery of specific knowledge domains (Cheng et al., 2019; Sun, Zhang et al., 2025; Wang, Gao et al., 2024). With the rapid development of intelligent education, the demand for cognitive diagnosis in personalized student assessment and ability enhancement has grown significantly (Liu, Zhang, Wang, Yu, & Li, 2023; Yao et al., 2023). By analyzing interaction data between students and exercises, CD not only models students' knowledge state with precision but also dynamically evaluates their learning progress. This approach effectively improves learning efficiency and helps students clearly understand their progress. Fig. 1(a) illustrates common application scenarios of cognitive diagnosis, which has been widely adopted in various fields, such as game design (Chen & Joachims, 2016), medical diagnosis (Guo, Li, Yu, & Haake, 2017), and education (Sun, Liu et al., 2025; Sun, Zhang et al., 2024).

Cognitive Diagnosis Models (CDMs) have undergone several stages of development, with their performance heavily reliant on the design of interaction mechanisms to capture the complex relationships between students and knowledge concepts (KCs). Although traditional Item Response Theory (IRT) (Lord, 1952) and its extended version, Multidimensional Item Response Theory (MIRT) (Ackerman, 2014), have provided a theoretical foundation for CDMs, their inherent assumptions present limitations in practical applications. As a classic form of CDMs, the DINA model (De La Torre, 2009) allows for an in-depth analysis of students' knowledge mastery, supporting the design of personalized learning pathways. In recent years, the integration of deep neural

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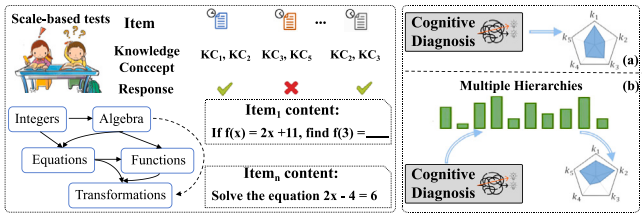


Fig. 1. A toy example of cognitive diagnosis. The upper part (a) depicts the traditional cognitive diagnosis approach, while the lower part (b) presents the proposed Multi-Hierarchy Interactive Constraint-aware Cognitive Diagnosis framework.

networks (Shen, Qian, Liu et al., 2024; Wang et al., 2020) and graph neural networks (Gao et al., 2021; Wang, Zeng, Yang, & Zhang, 2023) has enhanced model performance, paving the way for new directions in CD research.

Existing CD frameworks enhance diagnostic accuracy to some extent by analyzing students' explicit response records (e.g., exercise difficulty, related KCs) and specific behaviors (e.g., guessing behavior). However, these methods primarily focus on automatically inferring students' knowledge proficiency within the CD framework, failing to adequately consider the hierarchical relationships among different explicit records. As a result, they fall short of reasonably estimating students' true knowledge state. Introducing a hierarchical knowledge structure holds the potential to further improve diagnostic precision and interpretability, providing a more rational constraint on the estimation of students' mastery levels across various cognitive hierarchies, as illustrated in Fig. 1(b). Addressing this challenge will not only advance the field of cognitive diagnosis but also enhance the scientific rigor and fairness of modern educational assessment systems.

To more accurately diagnose students' knowledge proficiency, this study focuses on the key challenges of incorporating multi-hierarchy interactive constraint-aware features into the CD process. **First**, a fundamental prerequisite is how to reasonably define the hierarchy of explicit records and construct relationships between hierarchies in real-world educational contexts. **Second**, achieving effective feature interactions across multiple hierarchies to comprehensively represent both students' ability hierarchies and the hierarchies of item difficulty poses a critical challenge for improving model performance. **Finally**, students' knowledge proficiency is influenced not only by hierarchical features but also by individual differences. Therefore, enhancing CD through personalization and establishing intrinsic connections between individual and hierarchical features represent significant research directions and potential breakthroughs.

To address the aforementioned challenges, we propose a framework called the Multi-Hierarchy Interactive Constraint-aware Cognitive Diagnosis. This framework incorporates several key innovations aimed at enhancing the rationality and interpretability of cognitive diagnosis. **First**, we introduce a Hierarchical Relationship Construction Layer to identify hierarchical features of explicit records in real-world educational contexts and build associations between hierarchies. **Second**, we design a Multi-Hierarchy Interaction Awareness Layer to deeply analyze the interaction mechanisms between hierarchical features and implement an iterative constraint-aware strategy. **Finally**, we integrate a Diagnosis Enhancement Module, effectively combining multi-hierarchy interaction-aware features with personalized diagnostic features to enhance the expressive power of both individual and hierarchical features, enabling accurate inference of students' knowledge state.

In summary, we propose solutions to the identified challenges and, through this innovative approach, achieve more rational constraints on the dynamic changes in students' knowledge state. The main contributions of this study are as follows:

- We propose a Multi-Hierarchy Interactive Constraint-aware Cognitive Diagnosis Framework,[1] effectively constraining the evolution of students' knowledge state to make it more rational and closely aligned with real-world educational contexts.
- We design a portable Multi-Hierarchy Interaction Awareness Module that can be seamlessly integrated into existing cognitive diagnosis frameworks, significantly improving diagnostic accuracy.
- We combine hierarchical features with personalized diagnostic capabilities, enhancing the interpretability and reasoning of cognitive diagnosis and improving the practical application of personalized assessment.

2. Related works

Accurately assessing students' knowledge state in real-world educational contexts is of significant importance. We will elaborate on relevant research from the perspectives of cognitive diagnosis and hierarchical structuring in education.

2.1. Cognitive diagnosis

Cognitive Diagnosis (CD) is an assessment method originating from educational psychology that aims to infer students' knowledge mastery levels based on their response performance. This approach is founded on the assumption that students' cognitive state remain relatively stable over a short period (Zhou et al., 2021), enabling analysis and modeling based on their historical response behavior. Existing CD methods primarily follow two technical pathways: one rooted in traditional psychometric theories and the other leveraging neural network technologies to achieve higher-level nonlinear modeling.

Psychometrics: Models based on psychometric theories, such as Item Response Theory (IRT) (An & Yung, 2014; Lord, 2012), Multidimensional Item Response Theory (MIRT) (Ackerman, 2014), and the Deterministic Input, Noisy "And" Gate Model (DINA) (Yamaguchi & Okada, 2020), describe students' ability characteristics by modeling latent factors. For example, IRT associates students' abilities with item attributes (e.g., difficulty and discrimination) to quantify response performance (Gyamfi & Acquaye, 2023; Stucky & Edelen, 2014), while MIRT extends this framework by modeling students' cognitive traits as multidimensional vectors, providing a more accurate representation of the multidimensional nature of complex tasks (Chalmers, 2012; Finch, 2010; Jewsbury & van Rijn, 2020). The DINA model uses binary vectors to characterize students' mastery of KCs (Chen, Liu, Culpepper, & Chen, 2021; De La Torre, 2009; Tian, Wang, & Hong, 2023; Wang, Zeng, Yang, Xu and Zhang, 2024), assuming that students must fully master all relevant KCs to achieve the highest response accuracy. While these models have achieved significant success in diagnosing individual abilities, they exhibit limitations in modeling multidimensional cognitive traits and capturing dependency relationships between KCs. **The Multi-Hierarchy Interactive Constraint Module of MHCD framework can be seamlessly integrated to enhance psychometric diagnostic performance.**

Neural Networks: To address the limitations of traditional CD, NCDM (Wang et al., 2020) refines the modeling of students' cognitive state and item characteristics with multidimensional parameters and leverages neural networks to capture complex nonlinear interactions in heterogeneous data, significantly enhancing the capabilities of CD (Cheng et al., 2019; Li et al., 2022; Sun, Liu et al., 2025; Yu et al., 2024). Building on this, additional parameters or innovative structures have been introduced to further improve the diagnostic

[1] To support reproducible research, the data and code have been made publicly available at: https://github.com/xinjiesun-ustc/MHCD, encouraging further innovation in this field.

performance and interpretability of these models (Li et al., 2025; Qian et al., 2024). For instance, the ACD model (Wang, Zeng et al., 2024) integrates affective awareness with cognitive state, utilizing neural networks to jointly model emotions and cognition for more comprehensive diagnosis. The FairCD framework (Zhang et al., 2024) decomposes student abilities into fair and biased components via neural networks, effectively mitigating the influence of sensitive attributes on diagnostic results and enhancing fairness. These approaches, with their innovative neural network architectures, have significantly expanded the application scenarios and dimensions of cognitive diagnosis. **Our MHCD framework extends NCDM by incorporating these advancements.**

In recent years, researchers have increasingly recognized that the relationships among learners, exercises, and KCs can be effectively modeled using graph structures, leading to the introduction of Graph Neural Networks (GNNs) (Gao, Wang, & Ji, 2018; He et al., 2020) into CDMS. These studies primarily utilize GNNs to optimize the representations of learners, exercises, and KCs (Liu, Shao, & Zhang, 2021; Meng, Chen, Yi, & He, 2022), or to propagate information by modeling the mastery influence relationships among KCs (Shen, Qian, Zhang and Zhou, 2024; Wang et al., 2023). In particular, SCD (Wang et al., 2023) and DCD (Chen et al., 2024) primarily focus on single-level cognitive diagnosis. SCD enhances the handling of sparse interaction records through self-supervised learning, while DCD optimizes diagnosis by separating and aligning conceptual factors. In contrast, MHCD adopts a multi-hierarchies modeling approach, integrating KCs, exercises, and student abilities to provide a more comprehensive representation of students' knowledge mastery. Additionally, for interactions among learners, exercises, and KCs, most studies still rely on deep neural networks. For example, the RCD model (Gao et al., 2021) models student-item-knowledge relationships through multi-round feature learning, effectively enhancing the applicability and diagnostic performance of GNNs in cognitive diagnosis. **Our MHCD framework also adopts the concept of GNNs to enable joint interactions across multiple hierarchies.**

2.2. Educational hierarchy

The concept of educational hierarchy holds a significant place in educational research. Over many years of development, it has become one of the core methods for formulating efficient educational strategies and assessing student learning outcomes (Dlamini & Gerber, 2023; Kaviyaraj & Uma, 2022; Kemp, Palmer, & Strelan, 2019). The theoretical foundation of this approach can be traced back to Bloom's Taxonomy, which was first introduced in 1956 and has since had a profound impact on both educational theory and practice (Huang & Qiao, 2024). Bloom divided educational objectives into three domains: cognitive, affective, and psychomotor skills, with each domain further subdivided into different hierarchies. As educational needs have evolved, Bloom's Taxonomy has undergone several revisions to optimize the original hierarchical structure, better aligning it with the demands of modern education (Barari, RezaeiZadeh, Khorasani, & Alami, 2022; Bloom, Engelhart, Furst, Hill, & Krathwohl, 1956). These hierarchical frameworks not only provide tool support for educators in setting clear teaching objectives but also play a vital role in instructional design and educational assessment (Banda et al., 2023), thereby driving the ongoing development of educational theory and practice.

In practical application, hierarchical educational methods have effectively facilitated the precise setting of teaching objectives and the realization of personalized learning. Through hierarchical analysis, educators can design targeted teaching content based on students' individual needs and characteristics, significantly enhancing learning outcomes (Wang, Gao et al., 2024). For instance, Maslow's Hierarchy of Needs theory divides needs into different levels, revealing how needs at each level influence students' learning motivation (Yang et al., 2021). Gardner's Theory of Multiple Intelligences advocates for the cultivation of various intellectual abilities in students during the teaching process (Barari et al., 2022). Additionally, hierarchical frameworks have been widely applied in fields such as blended learning and programming education, where methods like Bloom's Taxonomy have improved students' learning achievements (Adijaya, Widiana, Agung Parwata, & Suwela Antara, 2023; Pikhart & Klimova, 2019).

With technological advancements, machine learning techniques have also been incorporated into the hierarchical structuring of educational objectives, improving the efficiency of educational resource matching through automated labeling and classification (Huang & Qiao, 2024; Li, Pardos, & Ren, 2024). For example, the HCD framework performs preliminary hierarchy formation based on students' relative performance and combines data analysis to offer new perspectives in supporting personalized teaching and knowledge tracking (Sun, Liu et al., 2024). Sun, Zhang et al. (2024) utilizes the prior distribution of educational data for hierarchical classification and dynamically adjusts students' target hierarchies, enhancing the interpretability of knowledge state modeling. These hierarchical analysis-based technologies and methods provide both a practical foundation and theoretical support for improving educational quality and meeting diverse learning needs. **Our MHCD framework introduces hierarchies for student ability, exercises, and KCs, leveraging Bloom's six levels to enhance interpretability.**

3. Preliminary

In this section, we formally introduce the concept of Multi-Hierarchy Interactive Constraint-aware Cognitive Diagnosis and provide a detailed description of the key components of the MHCD framework. Table 1 lists all the mathematical symbols used in this paper.

3.1. Problem definition

In the Multi-Hierarchy Interactive Constraint-aware Cognitive Diagnosis framework, we define several core sets. Specifically, the student set $S = \{s_1, s_2, \ldots, s_n\}$ contains n students; the exercise set $E = \{e_1, e_2, \ldots, e_m\}$ includes m different exercises; and the knowledge concept set $KC = \{kc_1, kc_2, \ldots, kc_k\}$ consists of k different knowledge concepts. Additionally, the framework includes several hierarchy sets: the knowledge concept hierarchy set $CH = \{ch_1, ch_2, \ldots, ch_h\}$, which describes the hierarchy of KCs; the exercise hierarchy set $EH = \{eh_1, eh_2, \ldots, eh_h\}$, which represents the hierarchy of exercises; and the student hierarchy set $SH = \{sh_1, sh_2, \ldots, sh_h\}$, which represents the students' ability performance at different hierarchies, h denotes the number of hierarchy levels involved in the framework, typically determined by the knowledge concepts, exercises, and the students' cognitive development levels. In this paper, we assume that all hierarchy sets have the same number of hierarchies, denoted by h. Finally, the response set $R = \{0, 1\}$ represents the student's answer status, where 1 indicates a correct answer and 0 indicates an incorrect answer.

A student's test record $\mathcal{T}$ is defined as a tuple set (s, e, ch, eh, sh), where $s \in S$ represents a student from the student set, $e \in E$ represents an exercise from the exercise set, $ch \in CH$ denotes the hierarchy level of the KC, $eh \in EH$ represents the hierarchy level of the exercise, and $sh \in SH$ indicates the student's ability hierarchy. To describe the relationship between exercises and KCs, a pre-determined matrix Q is introduced, denoted as $Q = Q_{ij}^{m \times k}$, where m is the number of exercises and k is the number of KCs. $Q_{ij} = 1$ if exercise e_i is associated with knowledge concept kc_j; otherwise, $Q_{ij} = 0$. Based on these definitions, this paper proposes the specific framework for Multi-Hierarchy Interactive Constraint-aware Cognitive Diagnosis.

Multi-Hierarchy Cognitive Diagnosis (MHCD): Based on the student's test records $\mathcal{T} = \{(s_n, e_1, ch_1, eh_1, sh_h), (s_n, e_2, ch_2, eh_2, sh_h), \ldots, (s_n, e_g, ch_h, eh_h, sh_h)\}$, and the known Q-matrix, this framework aims to infer the student's ability levels on different KCs. By incorporating a multi-hierarchy interactive constraint-aware mechanism, the

Table 1
Mathematical symbol and descriptions.

Symbol	Description
S, E	The set of students and exercises
KC, R	The set of knowledge concepts and responses
CH, EH, SH	The sets of knowledge concept hierarchy, exercise hierarchy, and student ability hierarchy
Q-matrix	Expert-annotated relationship matrix between exercises and knowledge concepts
θ^{sp}, θ^{sh}	Personalized knowledge proficiency and hierarchical knowledge proficiency
$\mathbf{h}_p^{diff}$, $\mathbf{h}_h^{diff}$	Exercise difficulty and hierarchical difficulty of exercises
n, m, k, h	The numbers of students, exercises, knowledge concepts, and hierarchy levels
CH^t, EH^t, SH^t	The hierarchical features of knowledge concepts, exercises, and student abilities after the tth iteration

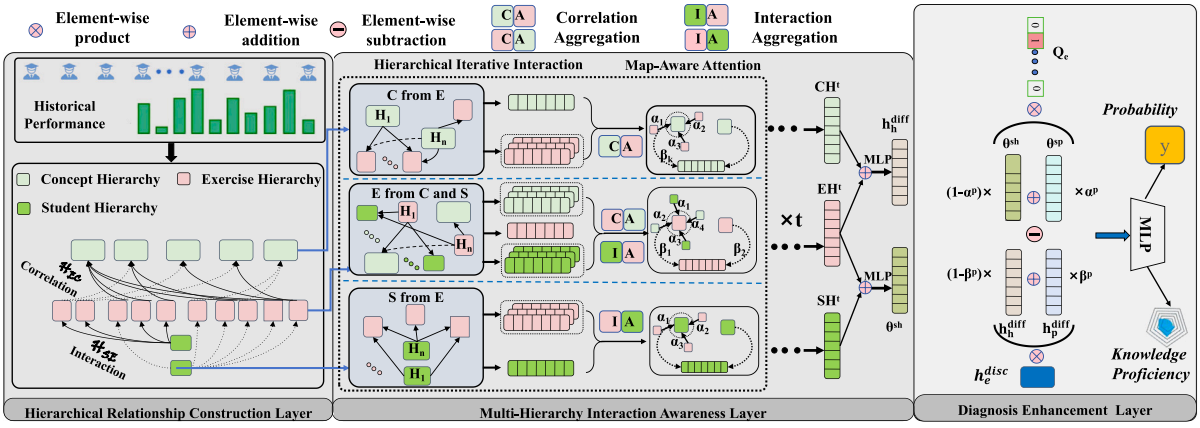


Fig. 2. The three core layers of the MHCD framework: Hierarchical Relationship Construction, Interaction Awareness, and Diagnosis Enhancement, where CH, EH, and SH represent the features of the Knowledge Concept Layer, Exercise Layer, and Student Ability Layer, respectively.

framework refines the student ability estimation process, making the evaluation results more accurate, reliable, and interpretable.

Hierarchy Constraint: In the MHCD model, structured limitations are imposed on the relationships among different hierarchies (knowledge concepts, exercises, and student abilities) to ensure model rationality and consistency. These include hierarchical consistency constraints (ensuring the alignment between knowledge, exercises, and student abilities) and taxonomy constraints (aligning with cognitive development hierarchies), which together enhance the model's interpretability and predictive capability.

3.2. Basic factors

Before conducting the interactive analysis in CD, it is necessary to explain the key foundational concepts to better understand the specific mechanisms involved in the interaction process.

The personalized knowledge proficiency vector of a student, denoted as $\boldsymbol{\theta}^{\mathbf{sp}}$, is a vector of length k, used to represent the student's proficiency across different KCs. The calculation process involves multiplying the student's one-hot encoded vector $\mathbf{x}_s$ by a trainable knowledge proficiency matrix $\mathbf{S}$, followed by mapping through a sigmoid activation function. The specific representation is as follows:

$$\boldsymbol{\theta}^{\mathbf{sp}} = \sigma(\mathbf{x}_s \times \mathbf{S}), \tag{1}$$

where $\boldsymbol{\theta}^{\mathbf{sp}} \in (0,1)^{1\times k}$ represents the personalized proficiency level of a student across all KCs. $\mathbf{x}_s \in \{0,1\}^{1\times n}$ is the one-hot encoding of the student, identifying the specific individual, and $\mathbf{S} \in \mathbb{R}^{n\times k}$ is a trainable parameter matrix that captures the relationship between students and KC proficiency levels. The function $\sigma(\cdot)$ denotes the sigmoid activation function, which normalizes the output values.

In the CD framework, the one-hot encoding $\mathbf{x}_e$ of exercises is used to represent exercise characteristics. Additionally, the exercise difficulty $\mathbf{h}_p^{\mathbf{diff}} \in (0,1)^{1\times K}$ and the exercise discrimination $\mathbf{h}_e^{\mathbf{disc}} \in (0,1)$ further describe the difficulty and discriminative power of the exercises. Their respective calculation formulas are:

$$\begin{aligned} \mathbf{h}_p^{\mathbf{diff}} &= \sigma(\mathbf{x}_e \times \mathbf{A}), \\ \mathbf{h}_e^{\mathbf{disc}} &= \sigma(\mathbf{x}_e \times \mathbf{B}), \end{aligned} \tag{2}$$

where $\mathbf{x}_e \in \{0,1\}^{1\times m}$ represents the one-hot encoding of exercise e, $\mathbf{A} \in \mathbb{R}^{M\times K}$ and $\mathbf{B} \in \mathbb{R}^{M\times 1}$ are trainable parameter matrices. Additionally, the Q-matrix is used to describe the relationship between exercises and KCs (Tatsuoka, 1983), and the calculation formula for its association vector is:

$$\mathbf{Q}_e = \mathbf{x}_e \times Q, \tag{3}$$

where $\mathbf{Q}_e \in \{0,1\}^{1\times k}$ reflects the degree of association between the exercise and the KCs.

4. The MHCD framework

In this section, we systematically present the proposed Multi-Hierarchy Interactive Constraint-Aware Cognitive Diagnosis Framework (MHCD) through the mathematical formulation of its cognitive functions. Subsequently, we delve into the key details and implementation mechanisms of each module within the framework.

4.1. Model overview

Typically, a student's performance on completing an exercise can be described using the following model:

$$r = \mathcal{M}(\boldsymbol{\theta}, \eta_e), \tag{4}$$

where r represents the student's response (e.g., correct answer or score), $\boldsymbol{\theta}$ denotes the student's knowledge proficiency, and η_e refers to exercise-related parameters (such as difficulty $\mathbf{h}^{\mathbf{diff}}$ and discrimination $\mathbf{h}_e^{\mathbf{disc}}$). The function $\mathcal{M}$ is typically based on a designed cognitive model, such as the interaction mechanisms in Neural Cognitive Diagnosis Models (NCDM). To ensure the rationality of student ability estimation,

considering that students typically exhibit higher similarity in performance within the same hierarchy, we introduce hierarchical ability constraints. These constraints capture the intrinsic relationships among students' abilities, thereby making the predicted results more reflective of their actual capabilities. Consequently, we decompose the student knowledge proficiency θ into two components:

$$\begin{aligned}\theta &= S(\theta^{\mathbf{sh}}, \theta^{\mathbf{sp}}),\\ \mathbf{h}^{\mathbf{diff}} &= D(\mathbf{h}_{\mathbf{p}}^{\mathbf{diff}}, \mathbf{h}_{\mathbf{h}}^{\mathbf{diff}}),\end{aligned} \tag{5}$$

where $\theta^{\mathbf{sh}}$ represents the ability characteristics constrained by the hierarchical structure, $\theta^{\mathbf{sp}}$ denotes the personalized ability characteristics of the student, $\mathbf{h}_{\mathbf{h}}^{\mathbf{diff}} \in (0,1)^{1\times k}$ indicates the exercise difficulty constrained by the hierarchical structure, and the functions S and D combine these two factors, reflecting the multidimensional influences of students' knowledge proficiency and exercise difficulty.

As shown in Fig. 2, the MHCD framework consists of three key stages: first, the establishment of hierarchical graph relationships; second, the multi-hierarchy interaction constraint perception modeling process; and third, the diagnostic enhancement stage. We will now describe the specific details and operational processes of these three stages.

4.2. Hierarchical relationship construction layer

Hierarchical information can provide valuable prior knowledge for the evaluation process (Vaughan, Lalonde, & Jenkins-Guarnieri, 2014). In the initial phase of multi-hierarchy interactive constraint-aware modeling, we constructed three types of hierarchical relationships based on students' historical performance: knowledge concept hierarchy, exercise hierarchy, and student ability hierarchy. To comprehensively capture the multi-hierarchy interaction between students, exercises, and KCs, we designed a graph structure where each node and edge can fully reflect the interactions and hierarchical features among these elements.

Firstly, the set of nodes in the graph consists of three types of nodes: exercises, KCs, and students, defined as:

$$V = E \cup KC \cup S \tag{6}$$

where E represents the set of exercises, KC represents the set of knowledge concepts, and S represents the set of students.

We use the relation matrices $R(e_i, k_j)$ and $R(s_i, e_j)$ to represent the associations between exercises and KCs, and between students and exercises, respectively. The specific expressions are:

$$\begin{aligned}\mathcal{E}_{ek} &= \{(e_i, k_j), (k_j, e_i) \mid R(e_i, k_j) = 1\},\\ \mathcal{E}_{se} &= \{(s_i, e_j), (e_j, s_i) \mid R(s_i, e_j) = 1\},\end{aligned} \tag{7}$$

where $R(e_i, k_j) = 1$ and $R(s_i, e_j) = 1$ indicate the construction of bidirectional edges to reflect the relationships between exercises and the KCs they involve, as well as the relationships between students and the exercises they have engaged with. These relationships effectively capture the bidirectional interactions and dependencies between exercises, KCs, and students.

We represent the hierarchical relationships of all nodes through a unified hierarchical identification function:

$$H(v_i) \leftarrow \mathcal{H}(v_i), \quad v_i \in \{e_i, c_j, s_i\}, \tag{8}$$

where $\mathcal{H}(v_i)$ is a unified hierarchical identification calculation function that computes the hierarchical identification for different types of nodes, specifically: $\mathcal{H}(e_i) = \ell_i^E$, $\mathcal{H}(c_j) = \ell_j^C$, and $\mathcal{H}(s_i) = \ell_i^S$, where the hierarchical identification for each type of node is computed based on the historical correctness rates of all students on exercises and knowledge concepts, as well as the historical correctness rates of all students on all items, with score intervals defined using the mean and standard deviation.

By combining the hierarchical identifiers of nodes and the relationships of edges, we finally construct two graph structures representing different hierarchical relationships: the KC and exercise hierarchy graph G_{ec}, and the exercise and student hierarchy graph G_{se}, represented as:

$$G_{ec} = (H_{ec}, \mathcal{E}_{ec}), \quad G_{se} = (H_{se}, \mathcal{E}_{se}), \tag{9}$$

where H_{ec} and H_{se} represent the hierarchical identifiers of the corresponding nodes, and $\mathcal{E}_{ec}$ and $\mathcal{E}_{se}$ represent the edge sets of the graphs, respectively depicting the complex hierarchical interaction relationships between KCs and exercises, and between exercises and students.

4.3. Multi-hierarchy interaction awareness layer

After constructing the hierarchical relationship graphs, we vectorize the representations of each hierarchy and adopt an iterative multi-hierarchy fusion approach to unify the modeling of interaction and structural relationships. Specifically, the interactive iterative learning is primarily based on the KC and exercise hierarchy graph G_{ec} and the exercise and student hierarchy graph G_{se}. Subsequently, we will elaborate on the update processes for the KC hierarchy, exercise hierarchy, and student ability hierarchy.

4.3.1. Knowledge concept hierarchy fusion

The embedding of the KC hierarchy $\mathbf{ch}_k^{(t)}$ is updated following a neighbor interaction rule based on an attention mechanism. First, the attention score $e_{kd}^{(t)}$ between node k and neighboring node d is calculated using the following formula:

$$e_{kd}^{(t)} = \xi\left(\mathbf{w}_a^{\top}\left[\mathbf{W}\mathbf{ch}_k^{(t)} \parallel \mathbf{W}\mathbf{eh}_d^{(t)}\right]\right), \tag{10}$$

where $\xi(\cdot)$ represents the LeakyReLU activation function, $\mathbf{eh}_d^{(t)}$ is the embedding representation of the exercise hierarchy, $\mathbf{w}_a$ is the learnable weight vector, and $\mathbf{W}$ is the weight matrix for the linear transformation. The symbol $\parallel$ represents the vector concatenation operation.

Subsequently, the calculated $e_{kd}^{(t)}$ will undergo normalization, resulting in the neighbor attention weight $\alpha_{kd}^{(t)}$, which is computed as follows:

$$\alpha_{kd}^{(t)} = \frac{\exp(e_{kd}^{(t)})}{\sum_{j\in\hat{G_{ec}}}\exp(e_{kj}^{(t)})}, \tag{11}$$

where $\hat{G_{ec}}$ represents the neighbor set of the current node k in the KC and exercise hierarchy graph G_{ec}.

Then, the information of the relevant exercise hierarchy neighbor nodes is aggregated with a weighted sum, using the attention weights $\alpha_{kd}^{(t)}$, to obtain the aggregated KC hierarchy embedding $\tilde{\mathbf{ch}}_k^{(t)}$:

$$\tilde{\mathbf{ch}}_k^{(t)} = \phi\left(\sum_{d\in G_{ec}} \alpha_{kd}^{(t)}\mathbf{W}\mathbf{eh}_d^{(t)}\right), \tag{12}$$

where $\phi(\cdot)$ is the Sigmoid activation function.

Finally, the KC hierarchy embedding is updated through multiple iterations using the following formula:

$$\mathbf{ch}_k^{(t+1)} = \mathbf{ch}_k^{(t)} + \beta_k^{(t)}\tilde{\mathbf{ch}}_k^{(t)}, \tag{13}$$

where the fusion weight $\beta_k^{(t)}$ is calculated using the following formula:

$$\beta_k^{(t)} = \frac{\exp(f_k)}{\sum_i \exp(f_i)}, \quad f_k = \mathbf{w}_b^{\top}[\mathbf{ch}_k^{(t)} \parallel \tilde{\mathbf{ch}}_k^{(t)}] + b, \tag{14}$$

where f_k is the score calculated through a linear transformation, and $\mathbf{w}_b$ and b are learnable parameters.

4.3.2. Exercise hierarchy fusion

To more comprehensively model the exercise embedding $\mathbf{eh}_d^{(t)}$, this paper iteratively updates it through the exercise-KC hierarchical graph G_{ec} and the exercise-student hierarchical graph G_{se}. The specific calculation process is as follows:

First, the exercise embedding is updated using the KC hierarchy, and the calculation formula is:

$$\tilde{\mathbf{eh}}_d^{(k,t)} = \phi\left(\sum_{k\in G_{ec}} \alpha_{dk}^{(t)} \mathbf{W}\mathbf{ch}_k^{(t)}\right), \tag{15}$$

where $\mathbf{ch}_k^{(t)}$ represents the current KC hierarchy embedding, G_{ec} is the set of neighboring KCs for exercise d, $\alpha_{dk}^{(t)}$ is the attention weight, representing the contribution of KC k to exercise d, and $\phi(\cdot)$ is the activation function.

Then, the exercise embedding is updated using the student hierarchy, and the formula is as follows:

$$\tilde{\mathbf{eh}}_d^{(s,t)} = \phi\left(\sum_{z\in G_{se}} \alpha_{dz}^{(t)} \mathbf{W}\mathbf{sh}_z^{(t)}\right), \tag{16}$$

where $\mathbf{sh}_z^{(t)}$ represents the ability embedding of student z, $\alpha_{dz}^{(t)}$ is the attention weight, and G_{se} is the set of neighboring students for exercise d.

Subsequently, the two update results are weighted and fused to obtain the final exercise embedding update:

$$\mathbf{eh}_d^{(t+1)} = \mathbf{eh}_d^{(t)} + \beta_1^{(t)} \tilde{\mathbf{eh}}_d^{(k,t)} + \beta_2^{(t)} \tilde{\mathbf{eh}}_d^{(s,t)}, \tag{17}$$

where the weight parameters $\beta_1^{(t)}$ and $\beta_2^{(t)}$ are computed adaptively, as follows:

$$\begin{aligned} &[\beta_1^{(t)}, \beta_2^{(t)}] = \frac{\exp(f_1)}{\sum_{i=1}^{2}\exp(f_i)}, \quad \frac{\exp(f_2)}{\sum_{i=1}^{2}\exp(f_i)}, \\ &f_1 = \mathbf{w}_c^{\top}[\mathbf{q}_d^{(t)} \parallel \tilde{\mathbf{q}}_d^{(k,t)}], \quad f_2 = \mathbf{w}_c^{\top}[\mathbf{q}_d^{(t)} \parallel \tilde{\mathbf{q}}_d^{(s,t)}], \end{aligned} \tag{18}$$

where $\mathbf{q}_d^{(t)}$ represents the current feature representation of exercise d, $\tilde{\mathbf{q}}_d^{(k,t)}$ and $\tilde{\mathbf{q}}_d^{(s,t)}$ are the updated embeddings of KCs and students, respectively, $[\cdot \parallel \cdot]$ denotes the feature concatenation operation, and $\mathbf{w}_c$ is a learnable weight.

Through the above update mechanism, the model dynamically captures the information of the exercise hierarchy in terms of the KC and student ability hierarchies, thereby enhancing the feature representation ability of the exercise hierarchy embedding. This multi-hierarchy fusion design not only strengthens the representation effect of the exercise embedding but also provides richer informational support for subsequent tasks.

4.3.3. Student ability hierarchy fusion

The update of student ability hierarchy embeddings is completed through message aggregation from exercise hierarchy information and dynamic adjustment of the embeddings themselves. The specific calculation is as follows:

First, based on the neighboring exercise set G_{se},

$$\tilde{\mathbf{sh}}_z^{(t)} = \phi\left(\sum_{d\in G_{se}} \alpha_{zd}^{(t)} \mathbf{W}\mathbf{eh}_d^{(t)}\right), \tag{19}$$

where $\alpha_{zd}^{(t)}$ is the attention weight, which measures the contribution of exercise d to the update of student z's embedding, and $\phi(\cdot)$ is the activation function.

Based on the message aggregation, the update formula for the student ability hierarchy embedding is:

$$\mathbf{sh}_z^{(t+1)} = \mathbf{sh}_z^{(t)} + \tilde{\mathbf{sh}}_z^{(t)}, \tag{20}$$

where $\mathbf{sh}_z^{(t)}$ is the current hierarchy embedding of student z, and $\tilde{\mathbf{sh}}_z^{(t)}$ is the update obtained through exercise aggregation.

Through the above calculation, the student embedding can dynamically capture the ability information associated with the exercises, gradually optimizing its representation capability with-in the model, and thus more accurately reflecting the student's knowledge state and ability level.

4.3.4. Multi-hierarchy fusion

Through multi-hierarchy interaction perception, the deep feature representations of KC hierarchy, exercise hierarchy, and student hierarchy after t rounds of interaction have been obtained. Based on these features, we can calculate the exercise's hierarchy difficulty $\mathbf{h}_{\mathbf{h}}^{\mathbf{diff}}$ and the student's knowledge proficiency level hierarchy feature $\boldsymbol{\theta}^{\mathbf{sh}}$, as follows:

$$\begin{aligned} \mathbf{h}_{\mathbf{h}}^{\mathbf{diff}} &= \sigma\left(\mathbf{W}_1 \cdot (\mathbf{eh}_d^{(t+1)} \parallel \mathbf{ch}_k^{(t+1)}) + \mathbf{b}_1\right), \\ \boldsymbol{\theta}^{\mathbf{sh}} &= \sigma\left(\mathbf{W}_2 \cdot (\mathbf{eh}_d^{(t+1)} \parallel \mathbf{sh}_z^{(t+1)}) + \mathbf{b}_2\right), \end{aligned} \tag{21}$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are learnable weight matrices, $\mathbf{b}_1$ and $\mathbf{b}_2$ are bias terms, and the activation function $\sigma(\cdot)$ is the Sigmoid function.

4.4. Diagnosis enhancement layer

In this stage, we fuse the exercise difficulty hierarchy features, the student's knowledge proficiency hierarchy features, the exercise difficulty, and the student's personalized knowledge level. By designing an adaptive optimization-based personalized weighting strategy, we effectively combine the hierarchical features with the personalized features to generate a comprehensive student state representation:

$$\begin{aligned} \boldsymbol{\theta} &= \alpha^{\mathrm{p}} \cdot \boldsymbol{\theta}^{\mathbf{sh}} + (1-\alpha^{\mathrm{p}}) \cdot \boldsymbol{\theta}^{\mathbf{sp}}, \\ \mathbf{h}^{\mathbf{diff}} &= \beta^{\mathrm{p}} \cdot \mathbf{h}_{\mathbf{h}}^{\mathbf{diff}} + (1-\beta^{\mathrm{p}}) \cdot \mathbf{h}_{\mathbf{p}}^{\mathbf{diff}}, \end{aligned} \tag{22}$$

where α^{p} and β^{p} are adjustable personalized weight coefficients that dynamically control the balance between the student's knowledge proficiency features and exercise difficulty features during the fusion process.

In this model, we adopt the basic paradigm of cognitive diagnostic models to predict the student's response to exercises. The specific formula is:

$$y = \mathbf{Q}_e \circ \left(\boldsymbol{\theta} - \mathbf{h}^{\mathrm{diff}}\right) \times h^{\mathrm{disc}}, \tag{23}$$

This method is applicable to various cognitive diagnostic models, such as IRT, MIRT, and NCDM.

4.5. Objective function

To optimize all the parameters in the MHCD framework, we process each response log record from the test data step by step and use the cross-entropy loss function to measure the difference between the predicted value y and the student's actual response label r. The specific form of the loss function is as follows:

$$\mathcal{L}(y,r) = -\sum_i \left(r_i \log(y_i) + (1-r_i)\log(1-y_i)\right) + \lambda \sum_j \theta_j^2 \tag{24}$$

where r_i represents the student's actual response label, and y_i represents the model's predicted probability. The term $\lambda \sum_j \theta_j^2$ is the L2 regularization, where λ is the regularization coefficient and θ_j represents the model's parameters. This regularization term penalizes large parameter values, helping to control the model's complexity and reduce the risk of overfitting. To minimize the above loss function, we use the Adam optimization algorithm for parameter optimization.

5. Experiments

In this section, we will introduce the real dataset used in the experiments and provide a detailed explanation of the key steps involved in the model training process, as well as the selected baseline models. Subsequently, we will focus on analyzing the improvement of the proposed MHCD framework in the student performance prediction task. To further explore the impact of the hierarchical constraints in the MHCD framework on model performance and knowledge state assessment, we will discuss the following core research questions:

Table 2
Statistics of all datasets.

Statistics \ Datasets	PISA-Science	PISA-Read	PISA-Math	Junyi
# of exercise records	7,585,349	811,136	2,594,902	353,835
# of correct answers	3,691,123	528,570	1,206,973	212,749
# of students	233,405	26,881	116,805	10,000
# of exercises	184	103	81	835
# of KCs	21	10	11	835
Average of exercise	32.50	30.18	22.22	35.38

- RQ1: What is the overall performance of the MHCD framework in student performance prediction?
- RQ2: Does the multi-hierarchy interaction constraint perception module in the MHCD framework exhibit good generalizability?
- RQ3: Can the MHCD framework effectively constrain and optimize student knowledge state representation?
- RQ4: What is the impact of varying the number of layers on model performance?
- RQ5: How interpretable is the MHCD framework in student performance diagnosis?
- RQ6: How does the MHCD framework improve student diagnostic outcomes?

Through in-depth investigation of the above questions, we aim to comprehensively validate the effectiveness and interpretability of the MHCD framework while exploring its potential application value in personalized education.

5.1. Datasets

To validate the performance of the multi-hierarchy interaction constraint perception cognitive diagnosis framework, we conducted experiments based on the PISA 2015 public dataset[2] and the Junyi dataset.[3]

- **PISA 2015.** The PISA 2015 dataset encompasses response records from over 500,000 students across 73 countries and regions. In this study, we selected sub-datasets from the domains of science, reading, and mathematics. During data preprocessing, responses labeled as "Full credit" were considered correct (label = 1), while those labeled as "No credit" and "Partial credit" were treated as incorrect (label = 0), and unanswered responses were ignored (Zhang et al., 2024). Additionally, to ensure the statistical significance of the samples, we retained only student records with no fewer than 30 responses in the science and reading datasets, while the threshold for the mathematics dataset was set to 20.
- **Junyi.** The Junyi dataset originates from Junyi Academy, an online learning platform in China that focuses on mathematics education. This dataset contains student learning logs collected between October 2012 and January 2015, along with detailed exercise-related information and annotations of exercise relationships. During preprocessing, we only considered each student's first submission for every exercise and retained only those students who had answered at least 15 questions to ensure statistical stability.

To ensure the reliability of the experimental results, all datasets were randomly divided into a training set (70%), a validation set (10%), and a test set (20%) to support model construction and performance evaluation. This rigorous data partitioning provides a solid foundation for the subsequent experiments. Table 2 summarizes the key statistical characteristics of each dataset.

5.2. Experimental setup

In the experiment, we first initialized the model parameters using the Xavier method and performed training through 5-fold cross-validation. The final evaluation results were based on the average of five rounds of validation. The experiment was run on a 64-bit Ubuntu 20.04.5 LTS server equipped with a 2.30 GHz Intel Xeon Gold 5218 processor and a 32 GB Tesla V100 GPU. The computation framework used PyTorch for model training and optimization.

5.3. Baselines

To comprehensively evaluate the performance of the MHCD model, we selected several classic baseline models for comparative experiments. All models were reproduced based on the public code repository[4] and fine-tuned through meticulous hyperparameter optimization to ensure optimal performance, thereby ensuring the fairness and reliability of the results. The experimental design used Area Under the Curve (AUC), Accuracy (ACC), and Root Mean Square Error (RMSE) as evaluation metrics to provide a multidimensional analysis of the model's performance.

- **IRT** (Lord, 2012): This method is based on Item Response Theory (IRT), modeling student performance to estimate their ability level and the difficulty of the questions. It is suitable for measuring a single latent ability.
- **MIRT** (Ackerman, 2014): MIRT is an extension of the IRT model that supports multi-dimensional latent ability modeling. It captures student responses across multiple dimensions while estimating both student abilities and item parameters, suitable for complex cognitive ability assessment.
- **DINA** (De La Torre, 2009): The DINA model assumes that each question involves multiple KCs and infers students' mastery of these concepts based on their responses. The model describes KC state in binary classification (ma-stered or not mastered), suitable for fine-grained cognitive diagnosis.
- **NCDM** (Wang et al., 2020): The NCDM combines neural networks to learn the complex relationships between students and items. Through a multi-layer network, this method efficiently performs CD and provides interpretable predictions of student abilities.
- **RCD** (Gao et al., 2021): The RCD framework enhances the accuracy and representation learning capability of CD by modeling the relationships between students, items, and KCs.
- **GCDM** (Su et al., 2022): The GCDM uses heterogeneous cognitive graphs to model interactions between students, skills, and items, integrating performance propagators and attention-based aggregators to assess skill mastery.
- **SCD** (Wang et al., 2023): The SCD method assists cognitive diagnosis through self-supervised learning and graph obfuscation techniques, providing more accurate diagnosis for students with sparse interaction records.

[2] https://www.oecd.org/en/about/programmes/pisa/pisa-data.html.
[3] http://www.junyiacademy.org/.
[4] https://github.com/bigdata-ustc/EduCDM.

Table 3
Results on student performance prediction across different datasets.

Methods	PISA-Science			PISA-Read			PISA-Math			Junyi		
	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE
IRT(Lord, 2012)	62.21	61.50	0.559	63.80	62.26	0.544	64.33	62.24	0.542	77.50	67.60	0.427
MIRT (Ackerman, 2014)	66.54	63.54	0.524	67.76	64.73	0.519	66.84	64.20	0.516	79.89	75.13	0.412
DINA (De La Torre, 2009)	67.67	64.83	0.547	68.92	65.71	0.539	68.15	65.47	0.533	73.5	67.9	0.463
NCDM (Wang et al., 2020)	78.91	71.18	0.434	79.21	74.38	0.417	80.61	72.69	0.424	76.14	71.38	0.445
RCD (Gao et al., 2021)	79.89	70.34	0.429	78.18	74.45	0.415	79.92	71.98	0.419	82.62	77.16	0.396
GCDM (Su et al., 2022)	78.35	70.92	0.437	79.01	74.10	0.423	80.15	72.12	0.428	81.55	76.34	0.401
SCD (Wang et al., 2023)	79.12	71.08	0.430	78.75	73.88	0.412	81.03	72.79	0.426	81.74	75.95	0.413
DCD (Chen et al., 2024)	78.45	70.87	0.433	79.55	74.05	0.414	81.52	73.65	0.412	80.11	75.21	0.414
ICDM (Liu, Shen, Qian, & Zhou, 2024)	77.98	70.45	0.437	78.92	73.60	0.419	80.08	72.34	0.425	79.58	73.25	0.432
MHCD (Ours)	**82.19***	**74.34***	**0.402***	**83.46***	**78.26***	**0.401***	**84.72***	**76.63***	**0.406***	**85.85***	**79.96***	**0.381***
+CHR	79.38	71.66	0.434	80.62	74.49	0.428	81.79	73.83	0.426	82.05	76.27	0.405
+EHR	78.91	71.14	0.442	80.21	73.92	0.435	81.43	73.51	0.435	82.61	76.46	0.402
+SHR	81.42	72.72	0.412	82.15	76.51	0.413	82.86	74.94	0.411	83.93	77.73	0.393

* Denotes a p-value less than 0.05 in t-test.

- **DCD** (Chen et al., 2024): The DCD model separates conceptual factors through students' response records and limited annotated information, aligning these factors to achieve efficient cognitive diagnosis.
- **ICDM** (Liu et al., 2024): The ICDM adopts a student-centered graph to inductively infer the knowledge mastery levels of new students without retraining the model, while maintaining high-performance diagnostic results.

5.4. Performance prediction (RQ1)

The true knowledge state of students cannot be directly measured, which makes performance evaluation of CD models challenging. Although improving the model's predictive accuracy is not our ultimate goal, student performance prediction serves as an indirect method for effectively testing the model's rationality. This analysis further highlights the unique value of the MHCD framework in capturing students' knowledge state.

Table 3 presents the experimental results of the MHCD framework, highlighting the following key points: First, the MHCD model achieves the best performance across all three major evaluation metrics, demonstrating that the multi-hierarchy joint modeling of KCs, exercises, and student abilities significantly enhances model performance. Second, compared to traditional methods, models based on neural cognitive diagnosis exhibit superior performance. For example, NCDM, RCD, and DCD outperform IRT and MIRT in predictive capability, with the RCD model, which leverages graph neural networks, further enhancing performance through its powerful feature learning ability. MHCD inherits this advantage and introduces a hierarchical node partitioning strategy based on KCs, exercises, and students, incorporating Bloom's six hierarchies of classification. This approach ensures efficient feature learning while effectively reducing the number of nodes. Compared to models like RCD, MHCD achieves significant optimization in computational complexity, with a complexity of $O(T\cdot(E\cdot d+6\cdot 3\cdot d))$, where E represents the number of edges, d is the embedding dimension, T denotes the number of iterations, and $6\cdot 3$ corresponds to the six hierarchical levels across three types of nodes. Third, whether on the large-scale PISA-Science dataset or the smaller PISA-Read dataset, the MHCD model demonstrates outstanding performance, showcasing its strong generalizability and further validating its adaptability and evaluation capabilities in large-scale data scenarios. Finally, even on the smaller Junyi dataset, the model maintains strong performance, indicating its robustness and stability.

In addition, to verify the effectiveness of the three key relational modules (CHR, EHR, and SHR) in the MHCD framework, we designed an ablation study. In the experiment, we retained each module individually while removing the others. Table 3 summarizes the experimental results under different conditions, leading to the following conclusions:

Table 4
Results on student performance prediction across different datasets.

Methods	PISA-Read			PISA-Math		
	AUC	ACC	RMSE	AUC	ACC	RMSE
IRT	63.80	62.26	0.544	64.33	62.24	0.542
MHCD-IRT	**67.12**	**65.88**	**0.522**	**68.15**	**67.34**	**0.519**
DINA	68.92	65.71	0.539	68.15	65.47	0.533
MHCD-DINA	**74.12**	**71.92**	**0.529**	**73.46**	**71.26**	**0.522**
RCD	78.18	74.45	0.415	79.92	71.98	0.419
MHCD-RCD	**79.25**	**74.99**	**0.413**	**80.51**	**72.55**	**0.414**

- Importance of each module: Regardless of which module is removed, model performance declines, indicating that all modules contribute significantly to the final performance and validating their effectiveness in modeling intra- and inter-hierarchical relationships.
- Key role of the SHR layer: The performance improvement is most significant when the SHR layer is used, indicating that properly utilizing the student ability hierarchy is crucial for accurately assessing students' knowledge state in multi-hierarchy interaction constraint modeling.

The above results thoroughly demonstrate the advantages of the MHCD framework in multi-hierarchy modeling and the indispensability of its key modules.

5.5. Transferability performance (RQ2)

The multi-hierarchy interactive constraint-aware module in MHCD demonstrates strong transferability while effectively capturing hierarchical features. To further validate its performance, we integrated this module into two classic traditional cognitive diagnosis frameworks (IRT and DINA) and one neural network-based cognitive diagnosis framework (RCD) and conducted experimental evaluations on two datasets. Table 4 summarizes the experimental results of these classic cognitive diagnosis models after incorporating the MHCD module.

Key conclusions from the experimental results: First, models incorporating the MHCD module (e.g., MHCD-IRT) significantly outperform their original counterparts (e.g., IRT and DINA) in terms of AUC, ACC, and RMSE, indicating that integrating multi-hierarchy constraint features into cognitive diagnosis effectively enhances model performance. Second, while applying the MHCD module to the RCD model achieves the best performance across all evaluation metrics, the improvement margin is relatively smaller compared to traditional models. This is likely because the RCD framework already leverages graph neural networks for comprehensive modeling. In contrast, traditional non-neural network models benefit more substantially from the deep

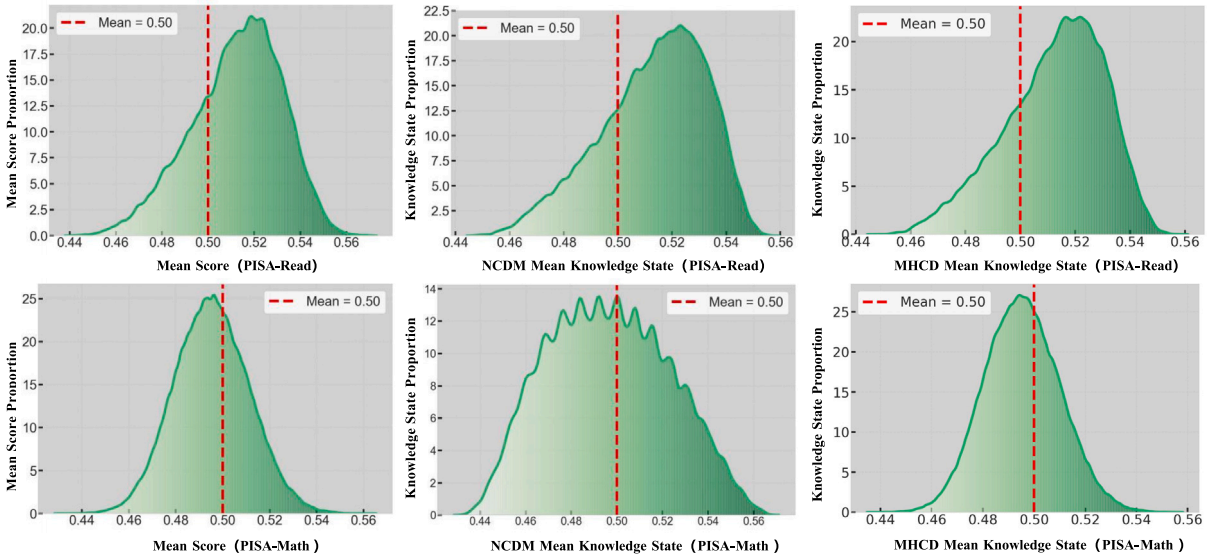


Fig. 3. Kernel density plots comparing the mean scores based on prior statistical data, the average knowledge state estimated by the NCDM model, and those predicted by the MHCD model across the PISA-Read and PISA-Math datasets.

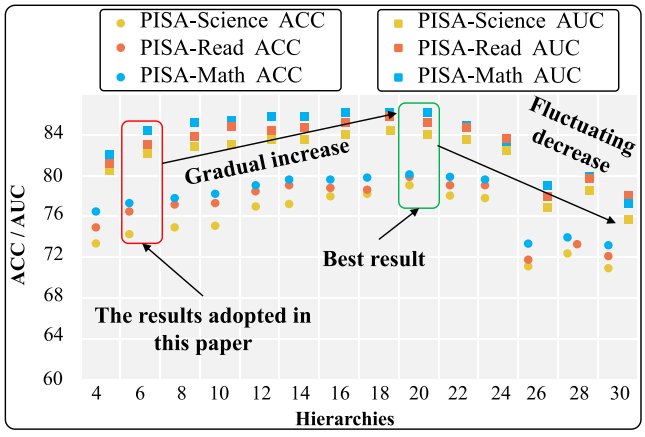


Fig. 4. The CEKT model tracks the evolution of a student's knowledge state over time as he/she solve three programming problems.

hierarchical feature capture enabled by the MHCD module, resulting in more pronounced performance improvements.

In conclusion, although the MHCD module exhibits strong generalizability and adaptability across frameworks, the experimental results highlight an important future direction: designing a CD framework specifically centered around the MHCD module. Such a framework could fully capitalize on the module's strengths, further improving the accuracy and effectiveness of diagnosing students' knowledge state.

5.6. Performance of hierarchy constraint aware (RQ3)

The introduction of multi-hierarchy interactive constraint awareness aims to enable cognitive diagnosis models to more accurately reflect students' abilities, thereby facilitating reasonable inferences about their proficiency across various KCs. This constraint helps ensure the rationality and validity of the assessment results. Fig. 3 illustrates the performance of multi-hierarchy interactive constraints on two typical datasets (PISA-Read and PISA-Math) through kernel density plots.

The experimental results reveal several key findings: First, both NCDM and MHCD (with multi-hierarchy constraints) effectively distinguish students' ability distributions across data. Their results align with the statistical trends of students' average scores, indicating that existing CD techniques demonstrate strong effectiveness in evaluating students' knowledge state. Second, a comparison of performance across the two datasets shows that while the knowledge state learned by both framework on the PISA-Read dataset exhibit average levels similar to students' prior distributions, their overall distributions deviate significantly from a normal distribution. In contrast, on the larger PISA-Math dataset, the distribution of student abilities is closer to a normal distribution, aligning with the general patterns of ability development. Third, the experiments indicate that on the PISA-Math dataset, the knowledge state learned by the NCDM model show greater fluctuations. However, after incorporating multi-hierarchy interactive constraints, the students' ability profiles become smoother and more closely aligned with the prior statistical distribution of evaluation scores. This finding suggests that the MHCD model not only captures more reasonable representations of students' ability state but also avoids sacrificing the rationality of ability representation merely to improve prediction accuracy, thereby enhancing the interpretability of the model.

Lastly, further analysis of both datasets, particularly focusing on the distribution of students with an ability level of 0.5, reveals that the model incorporating multi-hierarchy interactive constraint awareness achieves a distribution density closer to the prior distribution. For example, in the PISA-Math dataset, the prior distribution indicates that 24.0% of students have an ability level of 0.5, while the NCDM framework estimates only 13.5%. In contrast, the MHCD framework estimates this proportion to be 24.5%, which aligns more closely with the prior distribution. This result further validates the positive role of introducing hierarchy-based interactive constraint awareness in more reasonably reflecting students' knowledge state.

5.7. Impact of hierarchy on performance (RQ4)

To investigate the impact of hierarchy divisions on model performance, Fig. 4 presents the experimental results for divisions ranging from 4 to 30. The results indicate that as the number of divisions increases from 4 to 20, the model performance improves progressively, reaching its peak at 20 divisions. However, further increasing the number from 20 to 30 leads to fluctuations and a slight decline in performance. This suggests that finer-grained hierarchy divisions do not necessarily result in better performance. A possible explanation is that overly detailed divisions may increase data sparsity and amplify noise, negatively affecting the model's learning effectiveness and overall performance.

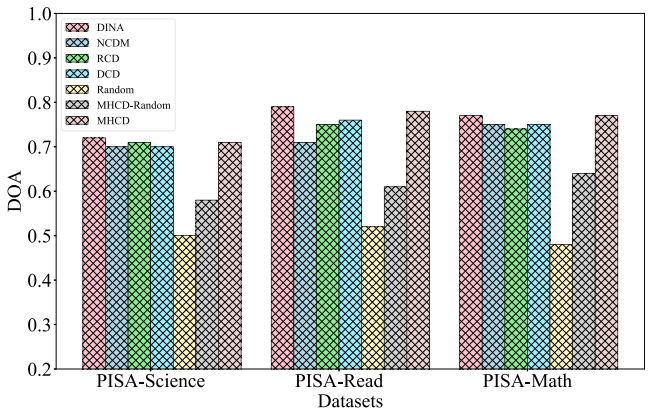


Fig. 5. Comparative analysis of model DOA interpretability across three distinct datasets.

Notably, inspired by Forehand et al. (2005) and Sun, Zhang et al. (2024), dividing the hierarchy into Bloom's six cognitive levels enhances interpretability. Therefore, while moderately increasing the number of divisions can enhance model performance to some extent, this study adopts a six-division hierarchy in the experiments to balance performance improvement with interpretability.

5.8. Interpretability of diagnostic results (RQ5)

The extent to which a student masters a specific knowledge component (KC) directly impacts their performance in problem-solving tasks. For instance, students with a higher level of mastery are more likely to excel in related tasks (Chen et al., 2017). To evaluate the model's effectiveness in capturing this relationship, we introduce the Degree of Agreement (DOA) metric (Wang et al., 2020). The DOA is defined as:

$$
\begin{aligned}
DOA(k) &= \frac{\sum_{a=1}^{N}\sum_{b=1}^{N} I(F_{ak}^{s} > F_{bk}^{s}) \cdot S_{abk}}{\sum_{a=1}^{N}\sum_{b=1}^{N} I(F_{ak}^{s} > F_{bk}^{s}) I(Z_0 > 0)}, \\
S_{abk} &= \frac{\sum_{j=1}^{M} I(Q_{jk} = 1) \wedge J(j,a,b) \wedge I(r_{aj} > r_{bj})}{Z_0}, \\
Z_0 &= \sum_{j=1}^{M} I(Q_{jk} = 1) \wedge J(j,a,b) \wedge I(r_{aj} \neq r_{bj}),
\end{aligned}
\tag{25}
$$

where F_{ak}^{s} represents the mastery level of student a on KC k; $I(\cdot)$ is an indicator function, where $I(X) = 1$ if the condition X holds true, and $I(X) = 0$ otherwise; Q_{jk} indicates whether exercise j is related to KC k; $J(j,a,b)$ verifies whether both students a and b answered exercise j; r_{aj} and r_{bj} represent the responses of student a and student b to exercise j, respectively; and finally, $I(r_{aj} > r_{bj})$ is used to check if student a's response is better than that of student b (see Fig. 5).

Traditional models (such as IRT and MIRT) describe student ability through latent trait vectors but fail to accurately reflect their mastery of specific KCs. The multi-hierarchy interactive constraint-aware model (MHCD) proposed in this study is compared with models such as Random, DINA, NCDM, RCD, and DCD, analyzing the differences between these models in terms of the DOA metric, which evaluates their ability to represent students' mastery of knowledge.

The analysis of the DOA results leads to the following key conclusions: First, the DINA model exhibits the best DOA performance across the three datasets, indicating its significant advantage in explaining students' knowledge mastery. This result confirms the theoretical basis of the DINA model, which relies on the explicit relationship between students and KCs. However, despite its strong interpretability, the DINA model still faces limitations in predictive capability in practical applications, as it fails to comprehensively capture the dynamic learning process of students. Secondly, neural network-based methods such as NCDM, RCD, and DCD perform well on the DOA metric, but still lag behind DINA in some aspects. The MHCD model, however, performs on par with DINA in terms of DOA, suggesting that the introduction of multi-hierarchy interactive constraints significantly enhances the interpretability of neural network-based models. Additionally, it is noteworthy that the DOA of the Random model, which assigns random values, stabilizes around 0.5, as expected. The MHCD-Random model, which integrates multi-level perception, shows a significant improvement in DOA, demonstrating that even in the absence of explicit mastery information, the perception mechanism can effectively enhance model performance. This phenomenon emphasizes the importance of multi-hierarchy perception learning in evaluating students' knowledge state and provides theoretical and practical foundations for improving model interpretability.

5.9. Case analysis (RQ6)

The relationship between students' cognitive diagnosis results, their abilities, and item difficulties can be quantified using Eq. (23). We randomly selected one student and their responses to 20 items from the PISA-Science dataset. The upper part of Fig. 6 displays the diagnostic results when the multi-level interactive perception mechanism is not introduced (i.e., the model degenerates into NCDM), while the lower part shows the diagnostic results using the MHCD framework. The following key conclusions can be drawn from the figure: First, both frameworks are able to effectively learn student abilities and item difficulties and generate reasonable prediction results based on these, indicating that both methods have reliability in modeling abilities and difficulties.

Second, the prediction match rate of the MHCD framework is 75.0%, significantly higher than the 60.0% of the NCDM framework. This suggests that MHCD performs better in terms of the accuracy of student performance prediction, being able to more precisely capture changes in students' knowledge state.

Finally, the interpretability score of MHCD is 85.0%, which is significantly higher than NCDM's 70.0%. In this study, interpretability is defined as follows: when a student's ability is greater than the item difficulty, the student should be able to answer the item correctly. If the prediction aligns with this principle, the diagnostic result is considered interpretable; otherwise, it is considered lacking in interpretability.

Specifically, for exercises 122 and 38, although both frameworks predict that the student will answer correctly, the actual results show that the student failed to answer correctly. In this case, MHCD provides a more reasonable explanation. For exercise 122, the ability learned by MHCD for the student is $\theta = 0.52$, and the item difficulty is $h^{\text{diff}} = 0.46$. Since the student's ability is greater than the item difficulty, the predicted probability of answering correctly is 0.67, which aligns with educational cognitive principles and demonstrates good interpretability. The prediction made by NCDM, on the other hand, does not satisfy this principle.

In contrast, for exercises 104 and 86, NCDM predicts that the student will answer correctly despite the student's ability being lower than the exercice difficulty, which contradicts educational cognitive principles. In comparison, MHCD accurately recognizes the student's insufficient ability and avoids erroneous predictions, further validating its advantages in interpretability.

In conclusion, the MHCD framework not only outperforms NCDM in terms of prediction accuracy but also demonstrates stronger interpretability through diagnostic results that align with educational cognitive principles. This highlights the significant value of the multi-hierarchy interactive perception mechanism in CD tasks.

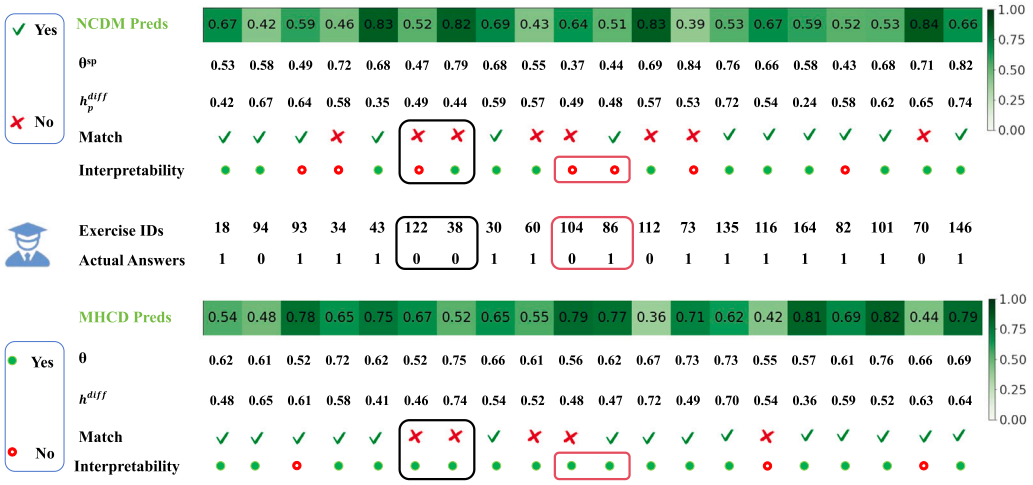


Fig. 6. MHCD cognitive diagnosis result example: Student ability θ_{sp}, student ability with integrated hierarchy θ, exercise difficulty h_e^{diff}, and exercise difficulty with integrated hierarchy h^{diff}.

6. Conclusions and limitations

In this paper, we proposed a novel Multi-Hierarchy Interactive Constrained Perception Cognitive Diagnosis framework (MHCD) aimed at accurately quantifying students' cognitive state. First, we constructed a hierarchical relational graph of KCs, exercises, and student abilities based on students' historical performance data, capturing the hierarchical relationships and associations among these three dimensions comprehensively. Then, we designed a multi-iteration hierarchical perception neural network that progressively integrates features across different hierarchies by associating KCs with exercises and exercises with student abilities. This iterative process generates more expressive representations of exercise difficulties and student abilities. Finally, during the diagnosis phase, the framework combines the multi-hierarchy interactive constrained student ability features with personalized knowledge proficiency to achieve precise evaluations of students' knowledge state. It also incorporates item difficulties and their multi-hierarchy features to provide more comprehensive support for CD.

The experimental results demonstrate that the MHCD model not only exhibits outstanding performance in student performance prediction and knowledge state estimation but also shows significant interpretability and transferability. Additionally, we applied the hierarchical features from the MHCD framework to three specific models (MHCD-IRT, MHCD-MIRT, and MHCD-DINA) and conducted extensive experiments on real-world data, further validating its effectiveness. However, the current multi-level interactive constrained perception mainly focuses on modeling students at an overall hierarchical level. Future research needs to further explore more fine-grained knowledge proficiency modeling and the integration of complex educational psychology factors, providing important directions and challenges for future studies. In addition, continuously optimizing computational efficiency is essential to ensure the model trains and infers efficiently while maintaining high performance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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